

MAKERERE UNIVERSITY

COLLEGE OF COMPUTING AND INFORMATION SCIENCES
DEPARTMENT OF NETWORKS

COURSE: BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING

COURSE UNIT: ARTIFICIAL INTELLIGENCE

COURSE CODE: CSC 2114

COURSEWORK TITLE: IMPLEMENTING UNINFORMED SEARCH USING A CAMPUS USE CASE

GROUP: B2

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1. Problem formulation

Component	Type	In our program	Changed since Assignment 1?
1. Initial state	One place from the set of 18 places (14 schools, 4 landmarks)	• The source school the user types • Example: School of Law	No
2. Actions(s)	Function: place → set of actions	• One action Go(n) for every place n joined to s • Read from graph[s], which now holds the distance in metres next to each neighbour • The distance does not decide which actions are allowed • Neighbours are taken in alphabetical order • Example: Actions(School of Law) = {Go(Freedom Square), Go(School of Economics)}	Yes to include distances
3. Result(s, a)	Function: place and action → place	Result(s, Go(n)) = n	No
4. Goal test	True/False test on a place	• True if the place is the destination school (never a landmark) • Applied when a node is removed from the frontier	No
5. Path cost and step cost	Numbers (metres)	• Step cost of walking from s to n = the distance, graph[s][n] • Path cost of a route = its total metres • Always positive: every distance is a whole number above zero (smallest is 70 m)	Yes: before, every step cost 1

What the weights still ignore

- Slope (Makerere is hilly), crowds and time of day (for example lecture changeovers), and rain or mud
- Slope can be added by making uphill walks cost more.
- It is still a positive number read from the graph, so UCS and the search loop do not change

Why positive step costs

- Cost 0: walking A → B → A would cost nothing, so a search without reached could go round and round at the same cost and never reach the dearer places
- Negative cost: a place we had already finished with could later be reached more cheaply, so UCS could return a route that is not the cheapest
- A negative loop would make the cost fall for ever

State versus node

- In UCS from School of Law to the Built Environment, the place Main Library appears in two nodes: 520 m (via School of Economics, 400 m edge) and 270 m (via Freedom Square)
- UCS should keep the 270 m node; the 520 m node is out of date ("stale")

What IDS needs from a node

- Its depth, the number of hops from the start
- A node works it out when it is created: depth = parent's depth + 1, and the root has depth 0

What each algorithm minimises

- BFS: hops. UCS: metres. IDS: hops, like BFS; its cost in metres is whatever that route happens to cost

2. Design: what changed since Assignment 1

- **Graph and checker.** build_graph makes a dictionary of dictionaries (place → neighbour → metres). neighbours(graph, s) gives (neighbour, metres) pairs in alphabetical order and step_cost reads one distance. route_cost(graph, route) adds up the distances again, and an assert in run_algorithm checks that it equals the printed cost. All search code receives graph as an argument.
- **Node.** STATE, PARENT, ACTION, PATH-COST (now metres from the start) and a new DEPTH = parent's depth + 1. Hops printed = DEPTH of the goal node. solution() is unchanged.
- **Reached.** Now holds the cheapest node found so far for each place. The update rule is a plug-in: never_replace (BFS, DFS, IDS keep the first node) or replace_if_cheaper (UCS replaces only if the new cost is strictly lower).
- **Three frontiers** with the same methods (push, pop, is_empty, __len__, push_order): QueueFrontier (first in, first out), StackFrontier (last in, first out; children are pushed in reverse so the alphabetically first comes out first) and PriorityFrontier (a heap of tuples (path_cost, state_name, counter, node)).
- **UCS ties and stale nodes.** The heap gives the lowest cost first, then the alphabetically first place, then the earliest added. The counter also stops Python comparing two Nodes. A heap cannot remove an item from the middle, so a replaced node stays inside. When it comes out, the loop sees reached[node.state] is not node and throws it away without counting it as expanded. With reached off, nothing is thrown away.
- **IDS driver.** ids() calls the same search() with limit 0, 1, 2, … (never above places − 1), using a StackFrontier, never_replace, reached off and the cycle check on. A node at the limit is still goal-tested but not expanded. search

returns CUTOFF if the limit stopped any node and FAILURE if nothing was cut off (print “No route found”). Expansions are added up over all rounds.

- **Weights never steer BFS, DFS or IDS.** They are read only in step_cost, in the UCS heap and in route_cost. Test: setting every weight to 1 gave the same routes and expansion counts for BFS, DFS and IDS on all 182 ordered school pairs.

The generic search loop. The frontier, the reached rule, the cycle check and the depth limit are the only parts that change between algorithms (marked with <-). There is no “if algorithm ==” branch.

```
search(graph, source, goal, frontier, replace_rule, use_reached, cycle_check, depth_limit, cap):
  root = Node(source); frontier.push(root); reached = {source: root}     # reached only if use_reached
  while frontier is not empty:
    node = frontier.pop()              # <- FRONTIER: queue / stack / priority (BFS / DFS+IDS / UCS)
    if use_reached and reached[node.state] is not node: continue    # stale node: throw away, not counted
    if node.state == goal: return SOLUTION(node)               # goal test on REMOVAL
    if depth_limit is not None and node.depth >= depth_limit:
      cutoff = True; continue                          # <- DEPTH LIMIT (IDS)
    if expansions == cap: return CAP
    expansions += 1
    children = []
    for place in neighbours of node.state (alphabetical):
      if cycle_check and place is on the path root..node: continue        # <- CYCLE CHECK (IDS)
      child = Node(place, node, place, node.path_cost + step_cost(graph, node.state, place))
      if use_reached and place in reached and not replace_rule(child, reached[place]): continue
                                        # <- REACHED RULE
      children.append(child)
    for child in frontier.push_order(children):                # <- FRONTIER decides push order
      reached[child.state] = child; frontier.push(child)
  return CUTOFF if cutoff else FAILURE
```

Algorithm	Frontier	Reached rule	Cycle check / depth limit	Reached table
BFS	QueueFrontier	never_replace	none	on (off with --noreached)
DFS	StackFrontier	never_replace	none	on (off with --noreached)
UCS	PriorityFrontier	replace_if_cheaper	none	on (off with --noreached)
IDS	StackFrontier (in ids())	never_replace	cycle check on, limit 0, 1, 2, ...	off

3. Verification (B4)

3.1 Reference rows

The program gives all three reference results exactly (cost in metres / hops / nodes expanded):

Source → Destination	UCS	IDS (total expanded)	Matches?
School of Education → School of Economics	550 / 4 / 8	560 / 2 / 4	yes
CoCIS → School of Liberal and Performing Arts	820 / 4 / 13	840 / 3 / 11	yes
School of the Built Environment → School of Law	870 / 5 / 11	1170 / 5 / 32	yes

3.2 Watching reached change (--trace, Law → Built Environment)

```
[trace] remove School of Economics g=120
[trace] push Main Library g=520 (from School of Economics)
[trace] remove Freedom Square g=180
[trace] reached replace Main Library: g=520 -> g=270
[trace] push Main Library g=270 (from Freedom Square)
[trace] discard stale Main Library g=520 (reached holds g=270)
```

- UCS removed Economics (120 m) before Freedom Square (180 m), because 120 is smaller
- Economics reached Main Library over the long 400 m edge, so reached stored Main Library at 520 m
- Freedom Square then reached Main Library over a 90 m edge for 270 m, which is cheaper, so reached replaced the entry and a new node was pushed
- The old 520 m node could not be taken out of the heap; it was thrown away without being expanded when it came out
- The first node was not good enough because it was the first to reach the place, not the cheapest. The same happened to Margaret Trowell (1100 m replaced by 920 m via Engineering), which leads to the best 870 m route

3.3 Tree-like UCS (reached off)

Pair	UCS, reached on: expanded	UCS, --noreached: expanded	Max frontier (noreached)
School of Education → School of Economics	8 (cost 550)	35 (cost 550)	87
School of Statistics and Planning → School of Veterinary Medicine	17 (cost 1840)	10,000: cap reached, no route returned	20,270

- Without reached, every loop in the graph (for example Main Building–Freedom Square–Main Library, or CoCIS–Central Teaching Facility–Statistics and Planning) is walked again and again
- The same place is made many times through different routes, so the frontier explodes: 87 nodes for a 550 m trip, over 20,000 for the 1840 m trip
- It would still stop without the cap: every step costs at least 70 m, so a route costing more than the best cost C* needs more than C*/70 steps. Only a limited number of routes cost C* or less, and once the cheapest goal node reaches the front of the heap it is removed and returned
- For 1840 m that is up to 26 levels of branching, far more than 10,000 expansions

3.4 IDS without reached

- The cycle check refuses any child whose place is already on the path from the start, so every path is simple (at most 17 hops)
- Each round has a depth limit, and the driver never goes above places − 1
- What IDS gives up: with no reached table it expands the same place again every time a different route reaches it (and again in every round), so it does much more work (see Q4). It also cannot replace a node by a cheaper one

4. Results

- The 40-row table (ten pairs, each run with BFS, DFS, UCS and IDS, reached on, ALL mode) is in Appendix A. For pair p, the rows are 4p−3 (BFS), 4p−2 (DFS), 4p−1 (UCS) and 4p (IDS)
- Every cost was checked by the route_cost assertion. BFS and DFS ignore the weights, so their routes are the same as before

Conditions met (row of the first algorithm vs row of the second):

Condition	Pairs (rows)
UCS route cheaper than BFS (at least 3)	1 (3 vs 1), 2 (7 vs 5), 3 (11 vs 9), 4 (15 vs 13), 9 (35 vs 33), 10 (39 vs 37)
UCS route has more hops than BFS (at least 1)	2 (7 vs 5), 3 (11 vs 9), 4 (15 vs 13), 9 (35 vs 33), 10 (39 vs 37)
IDS route costs more metres than UCS (at least 2)	1 (4 vs 3), 2 (8 vs 7), 3 (12 vs 11), 4 (16 vs 15), 9 (36 vs 35), 10 (40 vs 39)
UCS expands more nodes than BFS (at least 1)	3 (11 vs 9), 4 (15 vs 13), 7 (27 vs 25)
UCS expands fewer nodes than BFS (at least 1)	1 (3 vs 1), 5 (19 vs 17), 10 (39 vs 37)
IDS expands at least 5 times as many nodes as BFS (at least 1)	2 (8 vs 5), 8 (32 vs 29), 9 (36 vs 33)
The two schools are directly connected (at least 1)	5 (rows 17–20), 7 (rows 25–28)
Fewest-hops distance is 6 or more (at least 1)	2 (6 hops, rows 5–8), 8 (9 hops, rows 29–32), 9 (8 hops, rows 33–36), 10 (9 hops, rows 37–40)

--verbose on the deepest pair (pair 10, Veterinary Medicine → Built Environment: fewest-hops distance 9, and the UCS route has 10 hops, the most in the table):

Algorithm	Max frontier size	Cost (m)	Hops	Exp.
BFS	4	2320	9	17
DFS	7	2080	10	12
UCS	5	2080	10	16
IDS	8	2320	9	76

IDS per round (limit: expanded): 0: 0, 1: 1, 2: 2, 3: 3, 4: 4, 5: 5, 6: 7, 7: 11, 8: 22, 9: 21. Total 76, final depth limit 9.

One terminal run of ALL (Built Environment → Law; typed: School of the Built Environment, School of Law, all; the place menu is left out):

```
Algorithm: BFS
Route: School of the Built Environment -> Margaret Trowell School of Industrial and Fine Art ->
    School of Liberal and Performing Arts -> Main Building -> Freedom Square -> School of Law
Cost (metres): 1170  Hops: 5  Nodes expanded: 12

Algorithm: DFS
Route: School of the Built Environment -> Margaret Trowell School of Industrial and Fine Art ->
    School of Liberal and Performing Arts -> Main Building -> Freedom Square -> School of Law
Cost (metres): 1170  Hops: 5  Nodes expanded: 9

Algorithm: UCS
Route: School of the Built Environment -> School of Engineering -> CoCIS -> Central Teaching Facility ->
    Freedom Square -> School of Law
Cost (metres): 870  Hops: 5  Nodes expanded: 11

Algorithm: IDS
Route: School of the Built Environment -> Margaret Trowell School of Industrial and Fine Art ->
    School of Liberal and Performing Arts -> Main Building -> Freedom Square -> School of Law
Cost (metres): 1170  Hops: 5  Nodes expanded: 32  Final depth limit: 5
```

5. Analysis

Q1. UCS optimality

- Two conditions:
  (i) Every step cost is above zero, so cost always grows along a route;
  (ii) The goal test is done when a node is removed, not when it is made
- A removed node has the lowest cost on the frontier, so no unexplored route can beat it. A node that was only just made may have come by a poor route
- Pair 3 (Education → Economics): UCS returns 550 m (Education → Main Library → Freedom Square → Law → Economics)
- A UCS that tested when making a node would expand Education, then Social Sciences (140 m), then Main Library (160 m), make Economics there at 160 + 400 = 560 m and stop: Education → Main Library → Economics, 10 m dearer
- The 550 m route only reaches Economics after Law (430 m) is expanded (confirmed with a generation-test version of the loop)

Q2. BFS stops being optimal

- Pair 1 (Built Environment → Law), Figure 1. Both routes have 5 hops
- BFS: Built Environment → Margaret Trowell (70) → Liberal (450) → Main Building (350) → Freedom Square (120) → Law (180) = 1170 m
- UCS: Built Environment → Engineering (90) → CoCIS (250) → Central Teaching Facility (130) → Freedom Square (220) → Law (180) = 870 m
- BFS only counts hops, so all 5-hop routes look equal to it. It keeps the first one it finds, and the Margaret Trowell branch is queued first (alphabetical order), even though it uses the 450 m and 350 m edges
- Pair 2 shows the other case: UCS is cheaper (1580 m vs 1790 m) with more hops (8 vs 6)

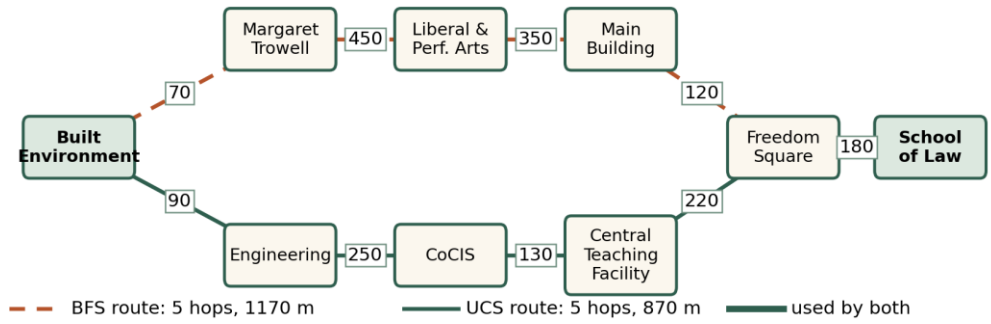


Figure 1. Built Environment → Law: BFS route (dashed) and UCS route (solid), with distances in metres.

Q3. IDS versus BFS and UCS

- IDS is never cheaper than UCS, because UCS is optimal. Same cost as UCS when the fewest-hops route is also the cheapest: pairs 5, 6, 7 and 8 (pair 8: 1990 m for both; pair 5: 100 m)
- Dearer when a route with more hops is shorter in metres: pair 2 (+210 m: 1790 vs 1580), pair 10 (+240 m), pair 1 (+300 m), pair 9 (+120 m), pair 3 (+10 m)
- In all ten pairs IDS costs the same as BFS, because both return the first fewest-hops route in alphabetical order

Q4. Where IDS's repeated work goes

- Pair 10, expansions at limits 0 to 9: 0, 1, 2, 3, 4, 5, 7, 11, 22, 21. Total 76, against 17 for BFS (about 4.5 times). Pair 8 is worse: 433 vs 17, with 183 at limit 8 alone
- Why it does not hold well here: the graph has only 18 places and an average of 2.7 neighbours, so the levels do not grow quickly
- With no reached table, IDS counts routes, not places. The number of simple routes through the loop-filled middle (Main Building, Freedom Square, Main Library, Central Teaching Facility, CoCIS, Statistics and Planning, Economics, Law) grows much faster than the 17 places BFS expands once each. The shallow rounds (55 of the 76 expansions) repeat all of it

Q5. Memory

- Max frontier size for pair 10: BFS 4 < UCS 5 < DFS 7 < IDS 8 (for IDS, the largest in any round). This is the opposite of the textbook order
- Why BFS stays small: the way from Veterinary Medicine through Agricultural Sciences, Food Technology and Natural Sciences to Education is a chain of places with only 2 neighbours. The frontier stays tiny until it reaches

the busy middle (Freedom Square and Main Library have 4 neighbours each), where each expansion adds up to 3 nodes

- BFS and UCS use reached, so each place is queued about once and the frontier cannot pass 18. UCS is one larger because its heap also holds stale entries (Main Building 1480, Economics 1700 and Statistics and Planning 1910 were replaced and later thrown away)
- DFS and IDS have no such limit: the stack keeps every unexplored neighbour along the current path (up to 9 deep), and IDS has no reached table

Q6. Two ways to avoid loops

Point	reached	Cycle check
Memory	One node for every place ever made (at most 18 here, more for a bigger map)	Nothing extra: it walks up the parent chain of the current path, taking time in proportion to the path length
Reopening a place at a lower cost	Only with the replace-if-cheaper rule, which lets UCS do it (Main Library 520 → 270 in 3.2)	Never: it does not compare costs
What it can wrongly throw away	• With never_replace: every later route to a known place, even a cheaper one (why DFS can return a dear route) • With a depth limit: a shorter route to a place first seen deeper (so IDS does not use it)	• Only routes that return to a place on their own path. With positive costs these are never better, so nothing useful is lost • Two different routes into the same place are not caught, which costs time

Q7. Do the uninformed algorithms read the weights?

- Prediction for Margaret Trowell–Liberal dropping from 450 m to 150 m: BFS, DFS and IDS never use weights to choose, so their routes and expansion counts cannot change, but a route that uses that edge is reported with a lower cost (cost is worked out again from the graph)
- UCS reads the weights, so both its route and its cost can change. Pairs whose routes avoid that edge are unaffected by all four
- Example, CoCIS → Liberal: BFS and IDS keep CoCIS → Engineering → Margaret Trowell → Liberal and its cost falls from 840 to 540. UCS switches from its 4-hop 820 m route via Freedom Square and Main Building to that same 3-hop route at 540 m (expansions 13 → 11)

Q8. Limitation of UCS

- UCS expands every place cheaper than the goal, in every direction
- Pair 6 (CoCIS → Social Sciences): 12 expansions, 4 of them moved away from the destination, meaning the place was farther from Social Sciences (true remaining metres) than the start was (670 m): Statistics and Planning (770), Engineering (920), Margaret Trowell (1000) and Built Environment (1010)
- A* (from the lecture's best-first table) would avoid most of this. It needs a guess h(n) of the distance left to the destination that is never too big, for example the straight-line distance in metres from map coordinates of every place, which our model does not hold yet. A* orders the frontier by g(n) + h(n)

6. Reflection

- Change: make the walking cost depend on direction. Walking uphill takes longer than walking down, so the same distance each way is unrealistic on Makerere's hills
- Component changed: only component 5 (the step cost differs in each direction), plus how the graph is stored (a directed dictionary). Everything else stays the same, and UCS stays correct while every cost is above zero. A time-of-day cost would be bigger: the state must include the clock time, which changes the state space, initial state and Result

Appendix A: results table (40 rows, graph search, reached on)

Key: MainBldg = Main Building; FreedomSq = Freedom Square; MainLib = Main Library; CTF = Central Teaching Facility; Liberal = School of Liberal and Performing Arts; SocSci = School of Social Sciences; MargTrowell = Margaret Trowell School of Industrial and Fine Art; BuiltEnv = School of the Built Environment; Eng = School of Engineering; Educ = School of Education; NatSci = College of Natural Sciences; FoodTech = School of Food Technology, Nutrition and Bioengineering; Agri = School of Agricultural Sciences; Vet = School of Veterinary Medicine; Law = School of Law; Econ = School of Economics; StatPlan = School of Statistics and Planning.

#	Source	Destination	Alg.	Route	Cost (m)	Hops	Exp.
1	BuiltEnv	Law	BFS	BuiltEnv → MargTrowell → Liberal → MainBldg → FreedomSq → Law	1170	5	12
2	BuiltEnv	Law	DFS	BuiltEnv → MargTrowell → Liberal → MainBldg → FreedomSq → Law	1170	5	9
3	BuiltEnv	Law	UCS	BuiltEnv → Eng → CoCIS → CTF → FreedomSq → Law	870	5	11
4	BuiltEnv	Law	IDS	BuiltEnv → MargTrowell → Liberal → MainBldg → FreedomSq → Law	1170	5	32
5	StatPlan	Agri	BFS	StatPlan → Econ → MainLib → Educ → NatSci → FoodTech → Agri	1790	6	16
6	StatPlan	Agri	DFS	StatPlan → CTF → FreedomSq → MainBldg → SocSci → Educ → NatSci → FoodTech → Agri	1860	8	12
7	StatPlan	Agri	UCS	StatPlan → CoCIS → CTF → FreedomSq → MainLib → Educ → NatSci → FoodTech → Agri	1580	8	16
8	StatPlan	Agri	IDS	StatPlan → Econ → MainLib → Educ → NatSci → FoodTech → Agri	1790	6	154
9	Educ	Econ	BFS	Educ → MainLib → Econ	560	2	7
10	Educ	Econ	DFS	Educ → MainLib → Econ	560	2	16
11	Educ	Econ	UCS	Educ → MainLib → FreedomSq → Law → Econ	550	4	8
12	Educ	Econ	IDS	Educ → MainLib → Econ	560	2	4
13	CoCIS	Liberal	BFS	CoCIS → Eng → MargTrowell → Liberal	840	3	11
14	CoCIS	Liberal	DFS	CoCIS → CTF → FreedomSq → MainBldg → Liberal	820	4	4
15	CoCIS	Liberal	UCS	CoCIS → CTF → FreedomSq → MainBldg → Liberal	820	4	13
16	CoCIS	Liberal	IDS	CoCIS → Eng → MargTrowell → Liberal	840	3	11
17	CoCIS	StatPlan	BFS	CoCIS → StatPlan	100	1	3
18	CoCIS	StatPlan	DFS	CoCIS → StatPlan	100	1	17
19	CoCIS	StatPlan	UCS	CoCIS → StatPlan	100	1	1
20	CoCIS	StatPlan	IDS	CoCIS → StatPlan	100	1	1
21	CoCIS	SocSci	BFS	CoCIS → CTF → FreedomSq → MainBldg → SocSci	670	4	12
22	CoCIS	SocSci	DFS	CoCIS → CTF → FreedomSq → MainBldg → SocSci	670	4	7
23	CoCIS	SocSci	UCS	CoCIS → CTF → FreedomSq → MainBldg → SocSci	670	4	12
24	CoCIS	SocSci	IDS	CoCIS → CTF → FreedomSq → MainBldg → SocSci	670	4	19
25	MargTrowell	Liberal	BFS	MargTrowell → Liberal	450	1	2
26	MargTrowell	Liberal	DFS	MargTrowell → Liberal	450	1	16
27	MargTrowell	Liberal	UCS	MargTrowell → Liberal	450	1	4
28	MargTrowell	Liberal	IDS	MargTrowell → Liberal	450	1	1
29	Eng	Vet	BFS	Eng → CoCIS → CTF → FreedomSq → MainLib → Educ → NatSci → FoodTech → Agri → Vet	1990	9	17
30	Eng	Vet	DFS	Eng → CoCIS → CTF → FreedomSq → MainBldg → SocSci → Educ → NatSci → FoodTech → Agri → Vet	2200	10	11
31	Eng	Vet	UCS	Eng → CoCIS → CTF → FreedomSq → MainLib → Educ → NatSci → FoodTech → Agri → Vet	1990	9	17
32	Eng	Vet	IDS	Eng → CoCIS → CTF → FreedomSq → MainLib → Educ → NatSci → FoodTech → Agri → Vet	1990	9	433
33	MargTrowell	Vet	BFS	MargTrowell → Liberal → MainBldg → MainLib → Educ → NatSci → FoodTech → Agri → Vet	2250	8	17
34	MargTrowell	Vet	DFS	MargTrowell → Eng → CoCIS → CTF → FreedomSq → MainBldg → SocSci → Educ → NatSci → FoodTech → Agri → Vet	2340	11	11
35	MargTrowell	Vet	UCS	MargTrowell → Eng → CoCIS → CTF → FreedomSq → MainLib → Educ → NatSci → FoodTech → Agri → Vet	2130	10	17
36	MargTrowell	Vet	IDS	MargTrowell → Liberal → MainBldg → MainLib → Educ → NatSci → FoodTech → Agri → Vet	2250	8	324
37	Vet	BuiltEnv	BFS	Vet → Agri → FoodTech → NatSci → Educ → MainLib → MainBldg → Liberal → MargTrowell → BuiltEnv	2320	9	17
38	Vet	BuiltEnv	DFS	Vet → Agri → FoodTech → NatSci → Educ → MainLib → FreedomSq → CTF → CoCIS → Eng → BuiltEnv	2080	10	12
39	Vet	BuiltEnv	UCS	Vet → Agri → FoodTech → NatSci → Educ → MainLib → FreedomSq → CTF → CoCIS → Eng → BuiltEnv	2080	10	16
40	Vet	BuiltEnv	IDS	Vet → Agri → FoodTech → NatSci → Educ → MainLib → MainBldg → Liberal → MargTrowell → BuiltEnv	2320	9	76

Appendix B: full --trace for UCS, School of Law → School of the Built Environment

```
[trace] remove School of Law g=0
[trace] push Freedom Square g=180 (from School of Law)
[trace] push School of Economics g=120 (from School of Law)
[trace] remove School of Economics g=120
[trace] push Main Library g=520 (from School of Economics)
[trace] push School of Statistics and Planning g=470 (from School of Economics)
[trace] remove Freedom Square g=180
[trace] reached replace Main Library: g=520 -> g=270
[trace] push Central Teaching Facility g=400 (from Freedom Square)
[trace] push Main Building g=300 (from Freedom Square)
[trace] push Main Library g=270 (from Freedom Square)
[trace] remove Main Library g=270
[trace] push School of Education g=430 (from Main Library)
[trace] remove Main Building g=300
[trace] push School of Liberal and Performing Arts g=650 (from Main Building)
[trace] push School of Social Sciences g=500 (from Main Building)
[trace] remove Central Teaching Facility g=400
[trace] push CoCIS g=530 (from Central Teaching Facility)
[trace] remove School of Education g=430
[trace] push College of Natural Sciences g=730 (from School of Education)
[trace] remove School of Statistics and Planning g=470
[trace] remove School of Social Sciences g=500
[trace] discard stale Main Library g=520 (reached holds g=270)
[trace] remove CoCIS g=530
[trace] push School of Engineering g=780 (from CoCIS)
[trace] remove School of Liberal and Performing Arts g=650
[trace] push Margaret Trowell School of Industrial and Fine Art g=1100 (from School of Liberal and Performing Arts)
[trace] remove College of Natural Sciences g=730
[trace] push School of Food Technology, Nutrition and Bioengineering g=1110 (from College of Natural Sciences)
[trace] remove School of Engineering g=780
[trace] reached replace Margaret Trowell School of Industrial and Fine Art: g=1100 -> g=920
[trace] push Margaret Trowell School of Industrial and Fine Art g=920 (from School of Engineering)
[trace] push School of the Built Environment g=870 (from School of Engineering)
[trace] remove School of the Built Environment g=870
```